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End Semester Examination 2024

Name of the Program: B. Tech. Biotechnology

Name of the Course: Himalayan Biodiversity and Biotechnology

Semester: VII

Course Code: TBT705c

Time: 3 hrs

Maximum Marks: 100

Note:-

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20 marks)	
(a)	Define the term <i>Indian Himalayan Region (IHR)</i> and identify the characteristics of the Greater and Lesser Himalayas. Provide an overview of their geographical features.	CO1
(b)	Explain the concepts of <i>Alpha</i> , <i>Beta</i> , and <i>Gamma</i> biodiversity with suitable examples. How do these levels contribute to understanding biodiversity in an ecosystem?	
(c)	Write short notes on the following concepts: (a) <i>Overexploitation</i> – What is it and how does it impact biodiversity? (b) <i>Biodiversity Conservation</i> – Why is it important and what are its key strategies?	
Q2	(20 marks)	
(a)	Name two bacteria found in hot springs. Explain how these extremophilic microorganisms are used in biotechnology. Provide practical examples of their applications.	CO2
(b)	Write a short note on (a) Microbial culture collection (b) bio-inoculants	
(c)	Define the term <i>Extremophiles</i> and explain with a suitable example. How do these organisms adapt to extreme environments?	
Q3	(20 marks)	
(a)	Name three medicinal plants found in the Indian Himalayan Region. Discuss their distribution and the medicinal properties they offer.	CO3
(b)	Explain the terms <i>In-situ</i> and <i>Ex-situ</i> conservation. How are these methods applied to conserve biodiversity in the Indian Himalayan Region?	
(c)	Assess the role of national parks and biosphere reserves in the conservation of biodiversity. What impact do these protected areas have on local ecosystems and species?	
Q4	(20 marks)	
(a)	Discuss the concept of <i>Wildlife Conservation</i> and analyze various strategies employed to protect wildlife. Provide examples of successful wildlife conservation initiatives.	CO4
(b)	How do pollinators contribute to biodiversity conservation? Discuss their role in ecosystems and agricultural systems with relevant examples.	
(c)	Write notes on the following topics concerning the Indian Himalayan Region. (i) Mammals biodiversity	

	(ii) Birds Biodiversity	
Q5	(20 marks)	
(a)	Justify the role of secondary metabolites in revolutionizing Ayurvedic medicines with the help of suitable examples.	CO5
(b)	Write short notes on the following medicinal plants from the Indian Himalayan Region: (i) <i>Picrorhiza kurrooa</i> (ii) <i>Aconitum heterophyllum</i>	
(c)	Defend the following statement with a suitable example- "Biotechnology as an important approach for the conservation of biodiversity"	